

Limits – 6

Asymptotics and sharp estimates: sums, integrals, Stirling

1 сентября 2026 года

Theoretical minimum.

Theorem 1 (sums against integrals). If f is monotone on $[1, n]$, then $|\sum_{k=1}^n f(k) - \int_1^n f(x) dx| \leq |f(1)| + |f(n)|$; for decreasing f the sequence $\sum_{k \leq n} f(k) - \int_1^n f$ is monotone and bounded, hence convergent. This gives $H_n = \ln n + \gamma + \frac{1}{2n} + O(\frac{1}{n^2})$.

Theorem 2 (Euler–Maclaurin, first step). $\sum_{k=a}^b f(k) = \int_a^b f(x) dx + \frac{f(a)+f(b)}{2} + \int_a^b \{x\}(1-\{x\}) \frac{f''(x)}{2} dx$ up to sign conventions; the further terms produce the Bernoulli numbers.

Theorem 3 (Stirling). $n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\theta_n}$, where $\frac{1}{12n+1} < \theta_n < \frac{1}{12n}$. The constant $\sqrt{2\pi}$ comes from Wallis' product.

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1. Prove Theorem 1 and the existence of Euler's constant γ ; obtain the first two terms of the asymptotics of H_n and estimate the error at $n = 10$.
 2. Prove Stirling's formula up to a constant by the sums-against-integrals method, and then determine the constant through Wallis' product.
 3. Let $S_n = \sum_{k=1}^n \frac{1}{\sqrt{n^2+k^2}}$. Prove that $S_n \rightarrow \ln(1 + \sqrt{2})$ and compute $\lim_{n \rightarrow \infty} n(S_n - \ln(1 + \sqrt{2}))$.
 4. Prove that there is a unique sequence $(c_m)_{m \geq 1}$ of numbers in $(0, 1)$ with $\int_0^1 \frac{dx}{1+x^m} = \frac{1}{1+c_m^m}$ for every positive integer m , and compute $\lim_{k \rightarrow \infty} k c_k^k$.
 5. Find the asymptotics of $\sum_{k=1}^n \sqrt{k}$ and of $\sum_{k=1}^n \ln^2 k$ up to $o(1)$, and say explicitly which Euler–Maclaurin terms you used.